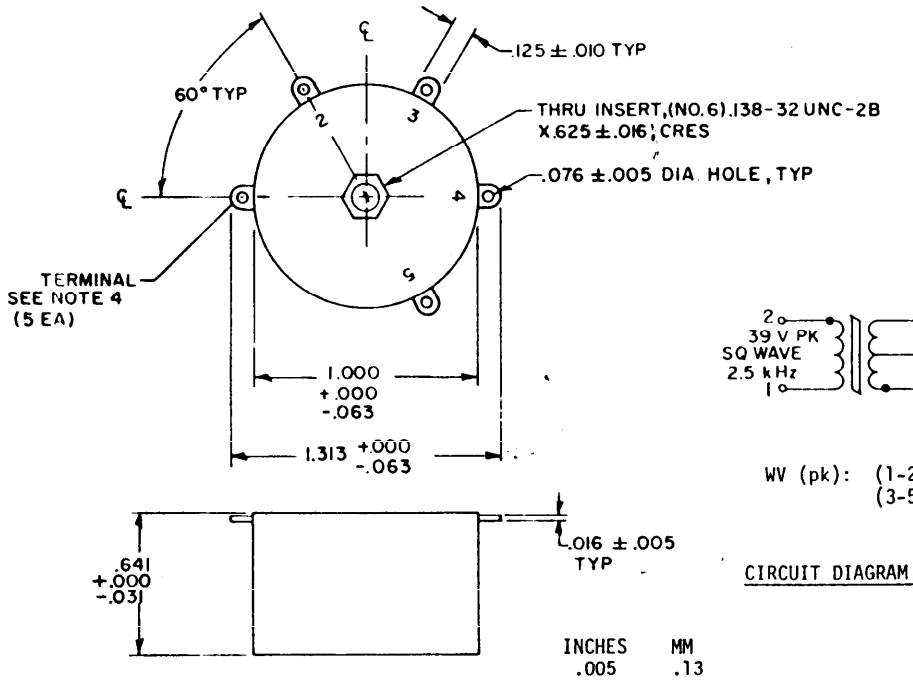
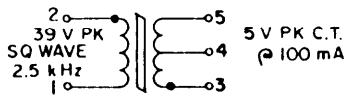
MILITARY SPECIFICATION SHEET

TRANSFORMER, SATURABLE, TF5S40ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.





WV (pk): (1-2) : 39 V
(3-5) : 5 V

CIRCUIT DIAGRAM AND MARKING

INCHES	MM
.005	.13
.010	.25
.016	.41
.031	.79
.063	1.60
.076	1.93
.125	3.18
.625	15.88
.641	16.28
1.000	25.40
1.313	33.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on side of case.
4. Terminals shall be brass, 70% Cu, 30% Zn, per Federal Specification QQ-B-613.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
Primary voltage (1-2): 39 volts peak, 2.5 kilohertz square wave.
Volt-ampere rating: 5.46 volt-amperes.
Secondary voltage (3-5): 5.0 volts peak ct.
Secondary current (3-5): 0.100 ampere average.
Working voltage (peak):
(1-2): 39 volts.
(3-5): 5 volts.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Terminals: Solder terminal.

Terminal height: $0.188\ ^{+0.000}_{-0.063}$ inch.

Weight: 56.7 grams, maximum.
Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 100 volts rms.

Electrical characteristics:
No load: With 39 volts peak, 2.5 kilohertz square wave across (1-2), current in (1-2) shall not exceed 16 milliamperes and the power in (1-2) shall not exceed 0.624 watt.
Voltage across (3-5): 5.25 volts ct ±3 percent.
Rated load: With 39 volts, 2.5 kilohertz square wave across (1-2), the voltage across (3-5) shall be 5.0 volts ct ±5 percent.
DC resistance:
(1-2): 9.01 to 12.19 ohms, maximum.
(3-5): 0.180 to 0.242 ohm, maximum.
Polarity: Additive with 2 and 5 connected.

Temperature rise: 50°C with 39 volts peak, 2.5 kilohertz square wave across (1-2) at an ambient temperature of 80°C.

Vibration (high frequency): MIL-STD-202, method 204, test condition A.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

Marking location: See figure 1.

Part number: M27/231-01.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AM, AR, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5950-0555-22)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-T-27/231 TRANSFORMER, SATURABLE, TF5S40ZZ

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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